

Desmond H. Grossmann, MSc, BSc

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🌐 [Website](#)



Summary

Final year PhD student specialising in **asteroseismology of red giant stars**, with significant modeling experience using **MESA** and **GYRE**. My research focuses on determining precise stellar ages from space mission data (*Kepler*, *TESS*, *Gaia*) and applying these constraints to questions of **galactic archaeology**.

Education

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| Nov 2023-early 2027 (est.) | 📖 Ph.D. in Astrophysics, Instituto de Astrofísica de Canarias & Universidad de La Laguna , Tenerife, Spain
Thesis working title: <i>Determining precise stellar ages for galactic archeology: Asteroseismic modeling of solar-like stars</i>
Thesis Supervisors: Savita Mathur & Paul Beck |
| Feb 2018 – Oct 2023 | 📖 M.Sc. Astrophysics, University of Graz & Graz University of Technology , Austria. (Part Time Studies due to higher education policy engagement)
Thesis title: <i>Asteroseismic Grid-modelling of the seismic-binary system KIC9163796</i>
Thesis Supervisors: Paul Beck & Arnold Hanslmeier |
| Aug 2016 - Dec 2016 | 📖 Exchange Semester, University of Calgary , Canada. |
| Oct 2014 – Feb 2018 | 📖 B.Sc. Physics, University of Graz & Graz University of Technology , Austria.
Thesis title: <i>KIC10614012 - A case study of an heartbeat eclipsing binary</i> .
Thesis Supervisors: Paul Beck & Arnold Hanslmeier |

Employment History

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| Oct 2023-early 2027 (est.) | 📖 Predocctoral Researcher, Instituto de Astrofísica de Canarias & Universidad de La Laguna , Tenerife, Spain |
| Feb 2023 – Sep 2023 | 📖 Assistant of the Vice Rector for International Affairs , University of Graz. ARQUS University Network co-coordination |
| Feb 2022 – Feb 2023 | 📖 Student assistant , Graz University of Technology. Responsible for COVID-19 reporting system. |
| Feb 2021 – Feb 2022 | 📖 Student assistant, EU Horizon Project "SOLARNET" , Organisation and development of the Summer School in "High Resolution Solar Physics", University of Graz. |
| Sep 2018 – Mar 2019 | 📖 Lecturer , Mechanics and Thermodynamics, Institute of Physics, University of Graz. |
| Oct 2016 – Jan 2021 | 📖 Teaching Assistant & Tutor, Institute of Physics, University of Graz . Assisting in various undergraduate courses (see teaching experience). |

Research Publications and Conference Contributions

The first author is the corresponding author. My full publication list is available in the public databases on [NASA ADS](#), or [ORCID](#). As by July 2026 I have an h-index of **h=5** from refereed publications. The current number citations per publications are given in brackets at the end of each entry.

Refereed Journal Articles – Submitted & Under Review

- 1 **Grossmann, D. H.**, Mathur, S., Beck, P. G., Godoy-Rivera, D., Liagre, B., García, R. A., Schimak, L. S., Palakkatharappil, D., Pinsonneault, M., and Pérez Hernández, F. (June 2026). “Ensemble Individual Mode Frequency Modeling of *Kepler* Giants”. In: *Astronomy & Astrophysics*, under review.
- 2 Godoy-Rivera, D., Richey-Yowell, T., Santos, A. R. G., Mathur, S., García, R. A., **Grossmann, D. H.**, and Beck, P. G. (2026). “Completing the magnetic puzzle: Photometric H-alpha as a stellar activity proxy”. In: *Frontiers in Astronomy and Space Sciences*, under review.
- 3 Rudrasingam, J., Lund, M. N., Grundahl, F., Chowhan, A., Bedding, T. R., Nissen, P. E., Serenelli, A., Ong, J. M. J., Li, Y., Panetier, E., Basu, S., Scutt, O. J., Metcalfe, T. S., Mathur, S., Winther, M. L., Hookway, G. T., Stokholm, A., **Grossmann, D. H.**, Larsen, J. R., Christensen-Dalsgaard, J., Huber, D., Fredslund, M. S., White, T. R., Beck, P. G., Pallé, P. L., García, R. A., and Kjeldsen, H. (2026). “Asteroseismology and interferometry of the F7V spectroscopic binary chi-Draconis A in the TESS CVZ”. In: *Astronomy & Astrophysics*, under review.

Refereed Journal Articles – Published & Accepted

- 1 **Grossmann, D. H.**, Beck, P. G., Mathur, S., Johnston, C., Godoy-Rivera, D., Zinn, J. C., Cassisi, S., Liagre, B., Masseron, T., García, R. A., Hanslmeier, A., Muntean, N., Schimak, L. S., Steinwender, L., and Stello, D. (Apr. 2025). “Improving the stellar age determination through joint modeling of binarity and asteroseismology: Grid modeling of the seismic red giant binary KIC 9163796”. In: *Astronomy & Astrophysics* 696, A42. [Citations: 9].  DOI: 10.1051/0004-6361/202452626. arXiv: 2501.09018 [astro-ph.SR].
- 2 Beck, P. G., **Grossmann, D. H.**, Steinwender, L., Schimak, L. S., Muntean, N., Vrad, M., Patton, R. A., Merc, J., Mathur, S., Garcia, R. A., Pinsonneault, M. H., Rowan, D. M., Gaulme, P., Allende Prieto, C., Arellano-Córdova, K. Z., Cao, L., Corsaro, E., Creevey, O., Hambleton, K. M., Hanslmeier, A., Holl, B., Johnson, J., Mathis, S., Godoy-Rivera, D., Simon-Díaz, S., and Zinn, J. C. (Feb. 2024). “Constraining stellar and orbital co-evolution through ensemble seismology of solar-like oscillators in binary systems. A census of oscillating red giants and dwarf stars in Gaia DR3 binaries”. In: *Astronomy & Astrophysics* 682, A7. [Citations: 44].  DOI: 10.1051/0004-6361/202346810. arXiv: 2307.10812 [astro-ph.SR].
- 3 Beck, P. G., Masseron, T., Pavlovski, K., Godoy-Rivera, D., Mathur, S., **Grossmann, D. H.**, Hamy, A., Palakkatharappil, D. B., Panetier, E., García, R. A., Merc, J., Lu, Y., Amestoy, I., and Deeg, H. J. (Feb. 2026). “Dynamical mass of a solar-like oscillator at the main sequence turnoff from Gaia astrometry and ground-based spectroscopy”. In: *Astronomy & Astrophysics* 706, L19. [Citations: 3].  DOI: 10.1051/0004-6361/202557452. arXiv: 2601.14197 [astro-ph.SR].
- 4 Schimak, L. S., Bedding, T. R., Crawford, C. L., Beck, P. G., Li, Y., Huber, D., Ong, J., Montet, B. T., Pedersen, M. G., **Grossmann, D. H.**, Mathur, S., and García, R. A. (Mar. 2026). “Testing red clump models with the asteroseismic binary KIC 10841730”. In: *MNRAS* 546.3, stag151. [Citations: 8].  DOI: 10.1093/mnras/stag151. arXiv: 2601.12773 [astro-ph.SR].
- 5 Godoy-Rivera, D., Mathur, S., García, R. A., Pinsonneault, M. H., Santos, A. R. G., Beck, P. G., **Grossmann, D. H.**, Schimak, L., Bedell, M., Merc, J., and Escorza, A. (Apr. 2025). “Kepler meets Gaia DR3: Homogeneous extinction-corrected color-magnitude diagram and binary classification”. In: *Astronomy & Astrophysics* 696, A243. [Citations: 23].  DOI: 10.1051/0004-6361/202348735. arXiv: 2501.18719 [astro-ph.SR].

- 6 Godoy-Rivera, D., Mathur, S., Richey-Yowell, T., Santos, A. R. G., Garcia, R. A., **Grossmann, D. H.**, Claytor, Z. R., and Beck, P. G. (July 2026). “Hints of enhanced magnetic activity after the intermediate rotation period gap as traced by the chromospheric Ca ii infrared triplet”. In: *arXiv e-prints*, arXiv:2607.05651. Accepted in Astronomy and Astrophysics, arXiv:2607.05651.  DOI: 10.48550/arXiv.2607.05651. arXiv: 2607.05651 [astro-ph.SR].
- 7 García, R. A., Mathur, S., Hookway, G. T., Godoy-Rivera, D., Masseron, T., Bétrisey, J., Buldgen, G., Lindsay, C., Metcalfe, T. S., Scutt, O. J., Stokholm, A., Beck, P. G., Benomar, O., Davies, G. R., Jiménez, A., Merc, J., Nielsen, M. B., Panetier, E., Pérez Hernández, F., Borg, L., Breton, S. N., Debacker, L., Escorza, A., and **Grossmann, D. H.** (June 2026). “Characterizing six seismic solar analogs observed by Kepler, K2, and HERMES”. In: *Astronomy & Astrophysics*, 710, A369. [Citations: 2].  DOI: 10.1051/0004-6361/202659849. arXiv: 2604.27090 [astro-ph.SR].
- 8 Liagre, B., García, R. A., Mathur, S., Pinsonneault, M. H., Serenelli, A., Zinn, J. C., Cao, K., Godoy-Rivera, D., Tayar, J., Beck, P. G., **Grossmann, D. H.**, and Palakkatharappil, D. B. (Oct. 2025). “Beyond the Nyquist frequency: Asteroseismic catalog of undersampled Kepler late subgiants and early red giants”. In: *Astronomy & Astrophysics* 702, A144. [Citations: 5].  DOI: 10.1051/0004-6361/202555167. arXiv: 2508.15654 [astro-ph.SR].

Non-refereed publications

- 1 **Grossmann, D. H.**, Mathur, S., Beck, P., García, R., Godoy-Rivera, D., Liagre, B., Muntean, N., and Palakkatharappil, D. (Oct. 2025). “Ensemble Boutique Modeling of Kepler Giants”. In: *TASC9/KASC16 9th TESS/16th Kepler Asteroseismic Science Consortium*, 109, p. 109.  DOI: 10.5281/zenodo.17433342.
- 2 **Grossmann, D. H.**, Beck, P. G., and Hanslmeier, A. (Jan. 2020). “KIC 9163796 - a benchmark binary for age determination”. In: *Stars and their Variability Observed from Space*. Ed. by C. Neiner, W. W. Weiss, D. Baade, R. E. Griffin, C. C. Lovekin, and A. F. J. Moffat, pp. 351–352.
- 3 Godoy-Rivera, D., **Grossmann, D. H.**, Richey-Yowell, T., Santos, A. R. G., Mathur, S., and García, R. A. (Mar. 2026). “Mining the Kepler Field: Atmospheric Parameters, Bolometric Corrections, and Luminosities”. In: *Research Notes of the American Astronomical Society*. Vol. 10. 3, 53, p. 53.  DOI: 10.3847/2515-5172/ae4fb6. arXiv: 2603.18152 [astro-ph.SR].
- 4 Schimak, L. S., Bedding, T., Crawford, C., Beck, P. G., Huber, D., Li, Y., Montet, B., Pedersen, M. G., and **Grossmann, D. H.** (Oct. 2025). “The asteroseismic binary KIC 1084173: What does it tell us about core helium-burning red clump star models?” In: *TASC9/KASC16 9th TESS/16th Kepler Asteroseismic Science Consortium*, 71, p. 71.  DOI: 10.5281/zenodo.17374474.
- 5 Beck, P. G., Kuschnig, R., Houdek, G., Kallinger, T., Weiss, W. W., Palle, P. L., Grundahl, F., Hatzes, A., Parviainen, H., Allende Prieto, C., Deeg, H. J., Jiménez, A., Mathur, S., Garcia, R. A., White, T. R., Bedding, T. R., **Grossmann, D. H.**, Janisch, S., Zaqarashvili, T., Hanslmeier, A., and Zwintz, K. (Jan. 2020). “The BRITE-SONG of Aldebaran - stellar music in three voices”. In: *Stars and their Variability Observed from Space*. Ed. by C. Neiner, W. W. Weiss, D. Baade, R. E. Griffin, C. C. Lovekin, and A. F. J. Moffat, pp. 75–79. arXiv: 2001.04912 [astro-ph.SR].

Invited Talks

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| 2026 |  Data Analysis Tools from Modern Space Telescopes for Use in the Classroom , IAU/NAEC organized Symposia „School and Space”, Klagenfurt, Austria, June 2026
 Asteroseismic grid-modeling of red giants: precise age constraints for galactic-archeology , The Good Vibrations seminar series, remote, May 2026 |
| 2024 |  Asteroseismology and the interior of stars , Dr.Hans-Riegel-Akademie, Brühl, Germany, September 2024 |

Invited Talks (continued)

- **Asteroseismic grid-modelling of the seismic-binary system KIC 9163796**, IMPRS PhD recruitment workshop, Max Planck Society, Garching, Germany, February 2023

Contributed Talks

- 2026 ■ **How Reliable Are the Asteroseismic Ages of Red Giants? Individual Mode Frequency Modeling from the Solar Regime to the Galactic Halo**, 10th TESS/17th Kepler Asteroseismic Science Consortium Meeting, Aarhus, Denmark, July 2026
- 2024 ■ **Asteroseismic grid-modelling of the seismic-binary system KIC 9163796**, European Astronomical Society Meeting, Session "Asteroseismology in multiple-star systems in the era of large photometric surveys", July 2024
- **Asteroseismic grid-modelling of the seismic-binary system KIC 9163796**, 8th TESS/15th Kepler Asteroseismic Science Consortium Meeting, Porto, Portugal, June 2024

Skills

- Languages ■ Strong reading, writing and speaking competencies for English and German, Basic communication abilities in Spanish, First Language: German
- Coding ■ Python, FORTRAN, MESA, SLURM (High Performance Computing), Astronomical Data Query Language (ADQL), SQL, $\text{\LaTeX}$, HTML
- Misc. ■ research experience, teaching experience, leadership

Awards & Achievements

As a primary applicant, I was awarded a total amount of **125.7k €** for my **research** and **studies** through awards and scholarships.

- 2023 ■ **"la Caixa" INPhINIT Doctoral fellowship**, (Full & independent PhD fellowship; 120k €, PI), awarded by the "la Caixa" foundation, Barcelona, Spain.
- 2022 ■ **Förderungsstipendium**, (Full cost scholarship for participation in "MESA Summer School 2022"; 2.0k €, PI), awarded by the University of Graz.
- 2018 ■ **Leistungsstipendium**, (Excellency Scholarship for outstanding academic performance; 0.8k €, PI) awarded by the Dean of Faculty of Natural Sciences, University of Graz.
- **Dr. Heinrich Jörg Stipendium**, (Conference Travel Scholarship 0.5k €) awarded by the Dr. Heinrich Jörg Foundation at University of Graz.
- **Participation and Travel Grant**, (for participation in "Dunlap Institute Summer School: Introduction to Astronomical Instrumentation"; 0.6k €, PI), awarded by the Dunlap Institute for Astronomy, University of Toronto, Canada (PI).
- 2016 ■ **Joint Study Stipendium**, (Scholarship for studying abroad; 1k €, PI), for exchange semester at the University of Calgary, Canada awarded by the Austrian Study Grant Authority.
- **KUWI-Stipendium**, (Scholarship for conference Participation, 0.5k €, PI), awarded by the University of Graz.
- 2014 ■ **Dr. Hans Riegel-Fachpreis**, (Award for excellency in final diploma school thesis, 0.3k €, PI) awarded by the Dr. Hans Riegel-Foundation, Bonn, Germany.

Teaching Experience

Independent Teaching Experience

- 2018-2019  **Lecturer in Exercise on Mechanics and Thermodynamics**, Bachelor's Course, 30 students, 3 ECTS-Credits, Senior Lecturer: Prof. Knoll.

Supervised Teaching and Tutoring

- 2024  **Tutor at "MESA Down Under" School**, Sydney, Australia, June 2024
- 2021-2022  **Vector Analysis**, Bachelor's Curriculum. Winter Term 2021, Supervisor: Prof. Helmut Gausterer.
- 2020-2021  **Advanced Physics Laboratory**, Bachelor's Curriculum. Winter Term 2020 and Winter Term 2021, Supervisor: Prof. Georg Koller.
- 2016-2018  **Quantum mechanics**, Bachelor's Curriculum, Summer Term 2018, Supervisor: Prof. Axel Maas. Theoretical Mechanics
-  **Beginner Physics Laboratory 1 & 2**, Bachelor's Curriculum, Summer Term 2017, Winter Term 2017, Summer Term 2018, Supervisor: Prof. Peter Knoll.

Service to the community

Academic recognition

- since 2026  **Refereeing for Astronomy & Astrophysics Journal**
- since 2025  **Co-Coordination of stellar research line meetings at Instituto de Astrofísica de Canarias**
- 2025  **Member of the Local Organisation Committee** of the TASC9/KASC16 Conference, July 2025, Vienna, Austria
-  **Jornadas de Puertas Abiertas del Observatorio del Teide (Outreach activity)**, June 2025, Tenerife
- 2024  **Member of the Local Organizing Committee** of the "SONG24 Science Meeting", September 18-20th 2024, Tenerife, Spain

Higher Education Policy Engagement

- since 2024  **Representative of PhD Students at Instituto de Astrofísica de Canarias**,
- 2022-2023  **Member of Audit Panel, Swiss Agency of Accreditation and Quality Assurance**, Audit and Re-certification of Vienna University of Technology (TU Vienna).
- 2022-today  **Steering Comitee & Student Council Member, ARQUS European University Alliance**, Project to stipulate European Universities by the European Union; nominated by the Rector of the Univesity of Graz.
- 2022- today  **Student's Member of the Academic Senate**, University of Graz & Graz University of Technology.
- 2019-2021  **Member of the Delegates Council, Austrian Science Fund (FWF).**
- 2019-2020  **Vice-President of Austrian Students' Union.** Statuary body, Nationwide Union of Students in Austria.
- 2018 - 2019  **Vice-President of the Student's Union at the University of Graz.**

Training Leads in Workshops and Seminars

- May 2022  **Higher Education Policy** for Student Representatives, Advanced Training Seminar by Students Union of Graz University of Technology, Söchau, Austria.